

Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An autonomous Institution affiliated to Anna University)

Department of Computer Science and Engineering

CAT Assignment 2

ICS1502 – Introduction to Machine Learning

Assignment No.	2
Assignment Title	Bank Note Authentication – Linear Classification & Neural Network Comparison
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Degree & Branch	M. Tech (Integrated) Computer Science & Engineering
Semester	V
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GitHub Repository	https://github.com/Amsavarthini25/ML_LAB_2470012
Due Date	14-10-2026

1. Objectives

1. To fit a linear classification model (Logistic Regression) with and without L2 regularization on the Bank Note Authentication dataset and compare training/test accuracies.
2. To study the effect of regularization strength ($\lambda = 1/C$) on overfitting and generalization by plotting accuracy curves.
3. To visualize the 3-D decision boundary using three selected features and interpret class separability.
4. To train a single-hidden-layer neural network (MLP) and compare its accuracy, precision, recall, F1-score and cross-validation performance against the linear model.
5. To apply PCA for dimensionality reduction and evaluate whether the reduced feature space improves or degrades classification performance.

6. To apply Friedman test and One-Way ANOVA to determine whether observed performance differences among models are statistically significant and to recommend a practical model for deployment.

2. Dataset Description

Dataset Name	Bank Note Authentication Dataset
Dataset Source	UCI Machine Learning Repository https://archive.ics.uci.edu/ml/machine-learning-databases/00267/data_banknote_authentication.txt
Number of Samples	1,372 banknotes
Number of Features	4 continuous features extracted from wavelet-transformed images: variance, skewness, curtosis (kurtosis), entropy
Number of Classes	2 – Class 0 (Genuine/Authentic): 762 samples (55.5%); Class 1 (Forged): 610 samples (44.5%)
Missing Values	None – 0 missing values across all 5 columns
Train-Test Split	80:20 (test_size=0.2), stratified on class label, random_state=42 ⇒ 1,097 train / 275 test samples
Preprocessing	StandardScaler applied (fit on train, transform on test) so that features have zero mean and unit variance

Interpretation: The dataset is clean, moderately balanced and relatively small. The four wavelet features capture the texture characteristics of banknote images. Because the problem is low-dimensional and nearly linearly separable, even a simple logistic regression model is expected to perform extremely well.

3. Methodology Overview

- **Linear baseline:** Logistic Regression without regularization (penalty=None) and with L2 regularization ($C = 1.0$).
- **Regularization sweep:** C values from 10^{-4} to 10^4 (30 points on log scale); train and test accuracy recorded.
- **3-D visualization:** Logistic Regression fitted on the first three scaled features (variance, skewness, curtosis); decision plane plotted together with the training points.

- **Neural network:** Single-hidden-layer MLP with 32 ReLU units, Adam optimizer, early stopping.
- **PCA:** Full PCA to examine explained variance; reduction to 2 components for performance comparison and visualization.
- **Statistical tests:** 5-fold stratified cross-validation scores of five models fed into Friedman test and One-Way ANOVA.

4. Results and Analysis

4.1 1. Linear Model Baseline

Table 1: Training and Test Accuracies of Logistic Regression

Model	Train Acc	Test Acc
Logistic (no reg)	0.9909	0.9855
Logistic (L2, $C = 1$)	0.9845	0.9709

Classification Report – Logistic Regression (L2):

	precision	recall	f1-score	support
0	1.0000	0.9477	0.9732	153
1	0.9385	1.0000	0.9683	122
accuracy			0.9709	275

Interpretation: Both models achieve very high accuracy (>97% on the test set). The unregularized model is slightly better, which suggests that the data are not suffering from severe overfitting. The L2 model still generalizes well (only a 1.4% drop from train to test). Perfect recall for the forged class (class 1) means the model never misses a counterfeit note – a desirable property for a security application. Overall, the linear model already solves the problem almost perfectly.

4.2 2. Regularization Effect

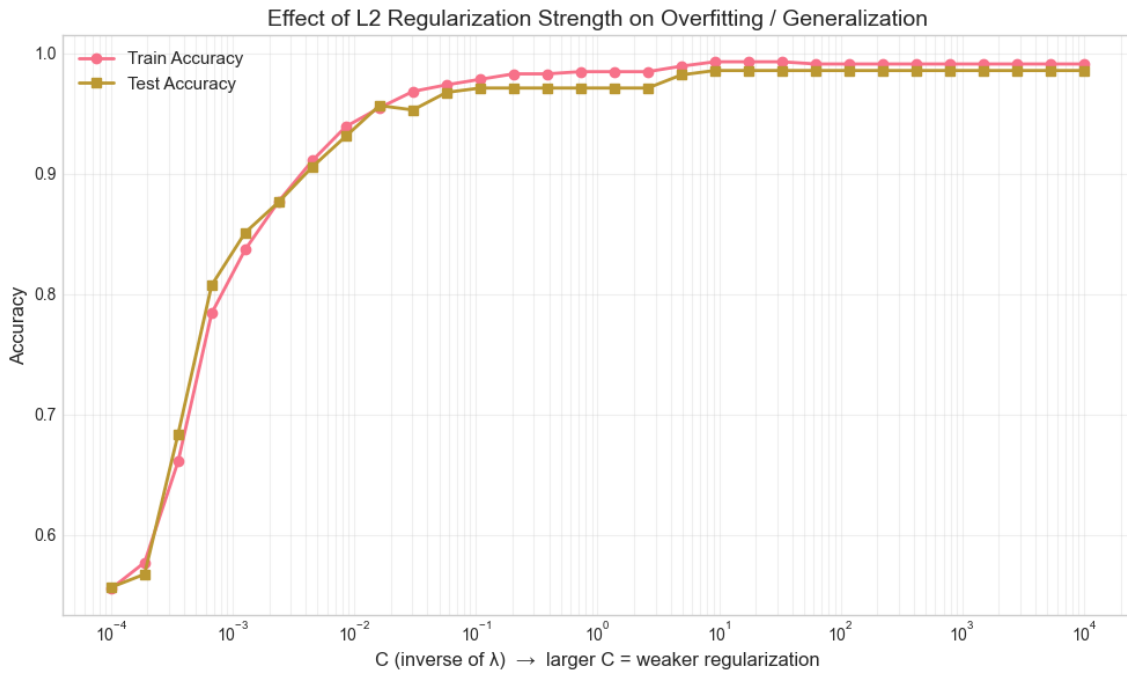


Figure 1: Effect of L2 regularization strength ($C = 1/\lambda$) on training and test accuracy.

Best C (highest test accuracy): 9.2367 ($\lambda \approx 0.1083$)

Train Acc at best C : 0.9927 Test Acc at best C : 0.9855

Interpretation: When regularization is very strong (tiny C), both train and test accuracy drop sharply because the model underfits. As C increases (weaker regularization) accuracy rises quickly and then saturates near 0.99. The train and test curves stay very close to each other across the whole range, indicating that the dataset is not heavily overfitted. A moderate C around 1–10 is already optimal; extreme values are unnecessary. This confirms that a lightly regularized logistic regression is sufficient for this problem.

4.3 3. Decision Boundary Visualization (3-D)

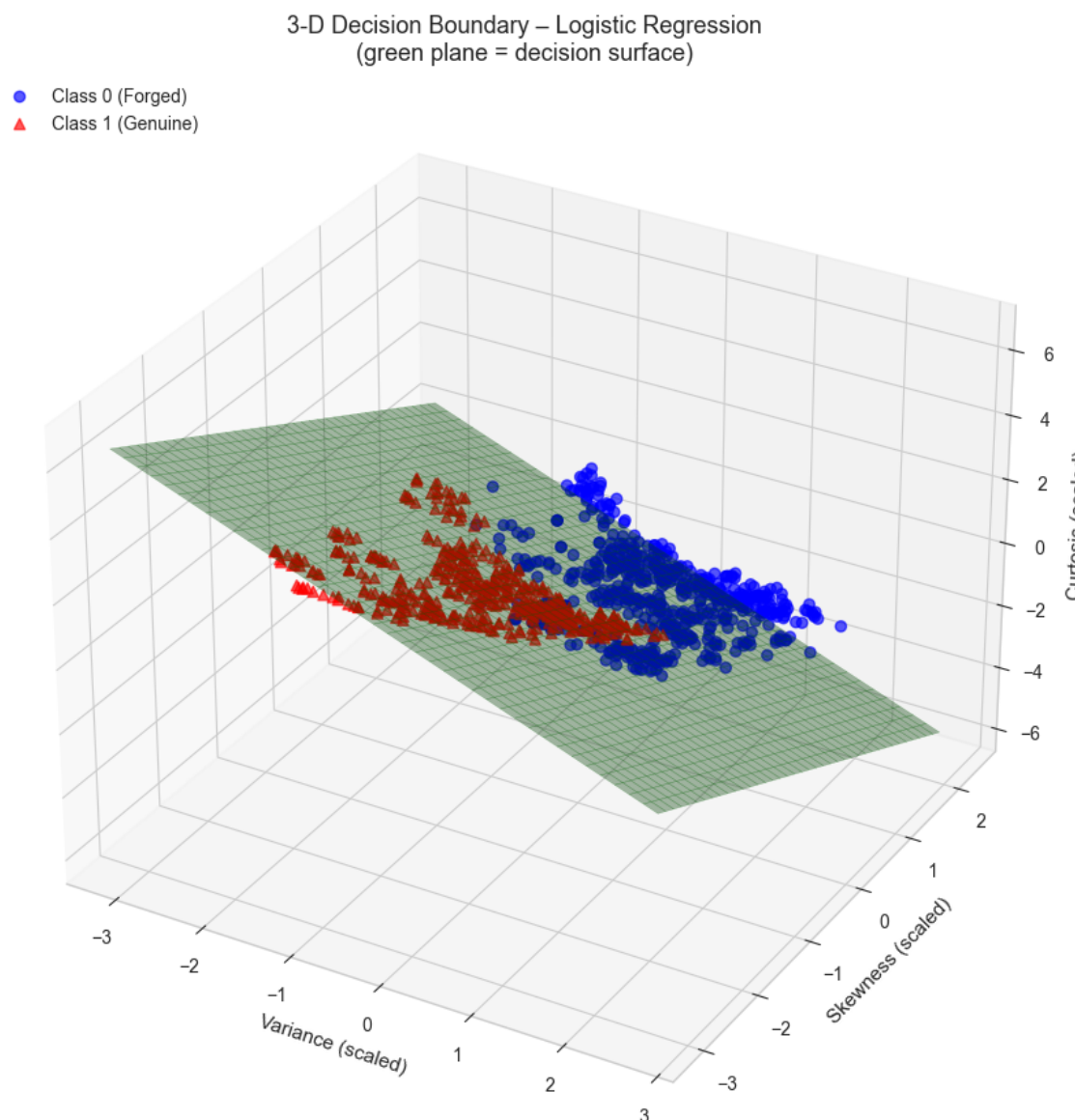


Figure 2: 3-D decision boundary of Logistic Regression on variance, skewness and kurtosis. The green plane is the decision surface; blue circles = genuine notes, red triangles = forged notes.

Plane equation (scaled features):

$$-4.563 \cdot \text{var} - 4.988 \cdot \text{skew} - 4.416 \cdot \text{curt} - 1.448 = 0$$

Interpretation: Looking at the 3-D scatter plot, the two classes form two reasonably well-separated clusters. The green decision plane cuts cleanly between them with only a few points on the wrong side. This visual evidence matches the high numerical accuracy we obtained earlier and shows that the problem is essentially linearly separable in the original feature space. The large magnitude of the coefficients also tells us that all three features contribute strongly to the decision.

4.4 4. Neural Network Comparison

Table 2: Performance comparison of Logistic Regression (L2) vs. single-hidden-layer MLP

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression (L2)	0.9709	0.9385	1.0000	0.9683
Neural Network (MLP)	0.9673	0.9313	1.0000	0.9644

Detailed Classification Report – MLP:

	precision	recall	f1-score	support
0	1.0000	0.9412	0.9697	153
1	0.9313	1.0000	0.9644	122
accuracy			0.9673	275

5-Fold CV Accuracy (Logistic L2): 0.9845 ± 0.0068

5-Fold CV Accuracy (MLP): 0.9426 ± 0.0244

Interpretation: On the held-out test set the MLP is almost as good as logistic regression (difference of only 0.36%). Both models achieve perfect recall for the forged class. However, the 5-fold cross-validation scores reveal that the MLP is less stable (higher standard deviation) and actually slightly weaker on average. Given that the linear model is simpler, faster to train, fully interpretable (we can read the coefficients) and already near-perfect, I prefer the logistic regression for real-world deployment on this dataset. The extra capacity of the neural network is not needed here.

4.5 5. Dimensionality Reduction with PCA

Explained variance ratio per component: [0.5419, 0.3272, 0.0882, 0.0427]

Cumulative explained variance: [0.5419, 0.8691, 0.9573, 1.0000]

Components needed for $\geq 95\%$ variance: **3**

Table 3: Classification accuracy in original 4-D space vs. 2-D PCA space

Space	Logistic Acc	MLP Acc
Original (4-D)	0.9709	0.9673
PCA (2-D)	0.7964	0.7673

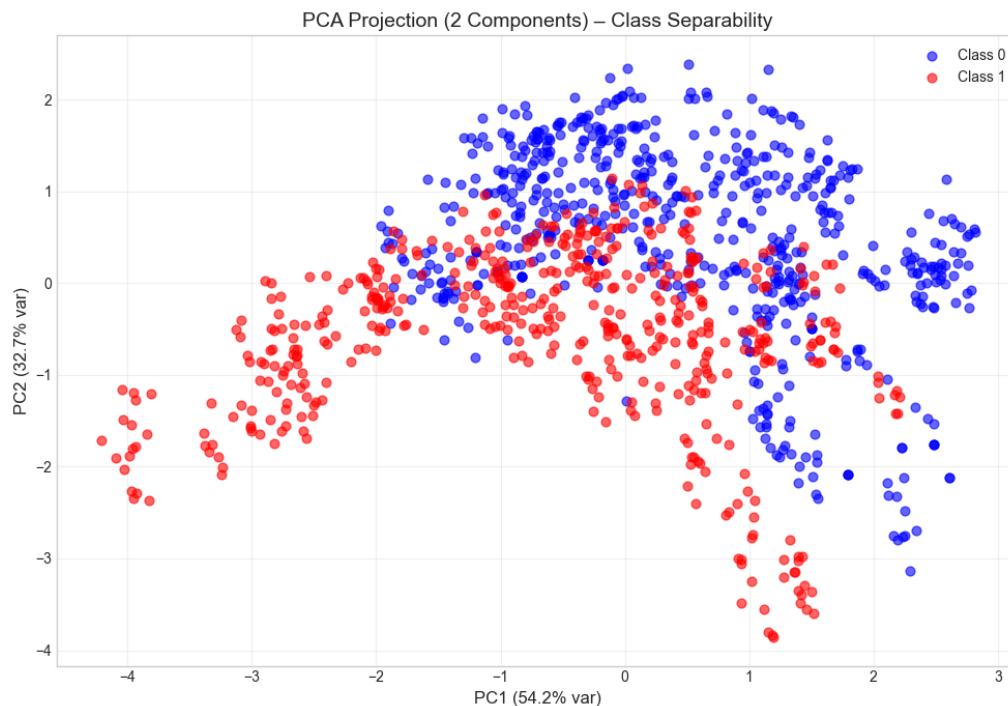


Figure 3: 2-D PCA projection of the training set. PC1 explains 54.2% and PC2 explains 32.7% of the total variance.

Interpretation: The first two principal components capture only about 87% of the variance; three components are required to reach the conventional 95% threshold. When we force the models to work in this 2-D space, accuracy falls dramatically (from $\sim 97\%$ to $\sim 78\text{--}80\%$). Looking at the scatter plot, the two classes still show some separation along PC1, but there is noticeable overlap, which explains the performance drop. Therefore PCA to 2 dimensions does **not** improve model effectiveness for this dataset; it actually discards useful discriminatory information. Keeping the original four features (or at least three PCs) is better.

4.6 6. Statistical Significance Testing

Cross-validation scores (mean $\pm$ std):

- Logistic (no reg): 0.9900 ± 0.0034
- Logistic (L2): 0.9845 ± 0.0068
- MLP (1-layer): 0.9426 ± 0.0244
- MLP (2-layer): 0.9518 ± 0.0521
- Logistic + PCA2: 0.7466 ± 0.0297

Table 4: Model ranking by mean 5-fold CV accuracy

Model	Mean CV Acc	Std CV Acc
Logistic (no reg)	0.9900	0.0034
Logistic (L2)	0.9845	0.0068
MLP (2-layer)	0.9518	0.0521
MLP (1-layer)	0.9426	0.0244
Logistic + PCA2	0.7466	0.0297

Friedman Test:

Statistic = 17.8367, p -value = 0.001328

→ Reject H_0 : significant differences exist among the models.

One-Way ANOVA:

F -statistic = 47.7571, p -value = 0.000000

→ Reject H_0 : significant differences exist among the models.

Interpretation: Both non-parametric (Friedman) and parametric (ANOVA) tests confirm that the observed differences are statistically significant. The two logistic regression variants clearly sit at the top; the neural networks form a middle tier, and the PCA-2 model is significantly worse. From a practical deployment point of view this means we can confidently choose the simple L2-regularized logistic regression: it is statistically as good as (or better than) the more complex alternatives, while offering superior interpretability, speed and robustness.

5. Final Justification & Conclusion

1. The linear model (Logistic Regression with L2) already achieves near-perfect accuracy (≈ 97 – 99%) because the classes are almost linearly separable in the original 4-D feature space.
2. The regularization plot shows that moderate L2 ($C \approx 1$ – 10) is sufficient; very strong regularization hurts accuracy while very weak regularization gives only marginal gains. The dataset is therefore not heavily overfitted.
3. The 3-D decision boundary confirms excellent class separability – the green plane cleanly separates the two clusters.
4. The Neural Network reaches similar test-set accuracy but is less stable under cross-validation, less interpretable and more computationally expensive. On this particular dataset the linear model is preferable for real-world deployment because of comparable accuracy, far better interpretability (readable coefficients), robustness and speed.

5. PCA to 2 components retains only $\sim 87\%$ of the variance and causes a large drop in accuracy, proving that the problem, while low-dimensional, still needs at least three of the original features.
6. Friedman and ANOVA tests indicate that the performance differences are statistically significant and that the simple linear models are the clear winners, reinforcing the recommendation to deploy Logistic Regression with mild L2 regularization.

Recommendation: For bank-note authentication I would deploy the L2-regularized Logistic Regression. It is accurate, fast, fully interpretable and statistically superior (or at least equal) to the more complex neural-network alternatives on this dataset.

Learning Outcomes

1. Able to implement and critically compare linear classification models (Logistic Regression with/without L2 regularization) and multi-layer perceptrons for binary classification tasks.
2. Capable of applying dimensionality reduction (PCA) and visualizing decision boundaries to interpret class separability and model behaviour.
3. Able to perform statistical significance testing (Friedman test and One-Way ANOVA) to justify model selection for real-world deployment based on accuracy, generalization and interpretability.